

# Yiğit Sarp Avcı

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## Education

**Boğaziçi University - B.Sc. in Computer Engineering**

09/2023 – Present

**Kabataş Boys' High School**

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

## Experience

**Executive Board Member at Radyo Boğaziçi**

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

**Work and Travel Program Participant**

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

**Regional Organizational Team Member at ESTIEM**

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

## Technical Projects

**AI-Powered FinTech Intelligence Platform** — Next.js, FastAPI, PostgreSQL, Docker

- Architected a full-stack, deployment-ready financial dashboard combining personal finance analytics, an algorithmic portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks.
- Engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine and achieved 100% backend test coverage with automated CI/CD pipelines.

**SchedSim CPU Scheduler** — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

**Spirelayer Command Interpreter** — C, Custom Backtracking Lexer

- Robust CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and complex command resolution.

**Aura Functional Interpreter** — Python, Functional Paradigms, Dual Scoping Engine

- Advanced interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines for complex variable resolution.

**MatrixNet** — Java, Tarjan's SCC Algorithm, Graph Theory

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection ( $O(V + E)$ ); optimized with Open Addressing to eliminate internal JVM string allocation overheads.

**GigMatch-Pro** — Java, Memory-Optimized Heaps

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for high-efficiency priority allocation and candidate-to-job matching algorithms.

**Nightpass Survival Game** — Java, Rule Engine Architecture

- Algorithmic implementation of a complex rule engine governing strategic deck, hand, and resource economics in a terminal-based survival card game environment.

## Volunteer Experience

**Education Mentor at TOÇEV**

09/2019 – 03/2020

- Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds.

Volunteer at Kabataş Boys’ High School Community Involvement Projects Club

09/2018 – 06/2023

- Carried out various volunteer projects aimed at helping and supporting people in need through community involvement initiatives.

Volunteering Division Member at Boğaziçi University Industrial Engineering Club

09/2023 – 06/2024

- Supported the organization and execution of volunteer initiatives and community-oriented projects within the club.

Education Mentor at Derece Kampüsü

- Mentored high school students for the university entrance exams, providing academic guidance and exam strategy support.

Technical Skills

- **Languages:** C, C++, Java, Python, TypeScript, JavaScript, x86-64 Assembly, VHDL, Bash
- **Web & Cloud Architecture:** FastAPI, Next.js, React, PostgreSQL, Docker, REST APIs, CI/CD
- **Low-Level & Systems:** Pthreads, Linux Syscalls, Memory Management, Process Scheduling, FPGA
- **AI & Modern Engineering:** Vibecoding, Autonomous Coding, LLM Prompting, AI-Assisted Architecture
- **Core Theory:** Data Structures & Algorithms, Complexity Analysis (Big O), Graph Theory, Discrete Math
- **Software & Ecosystem:** Linux/Unix, Git, GitHub, LaTeX, MS Excel, PowerPoint, Canva, Adobe Photoshop

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually